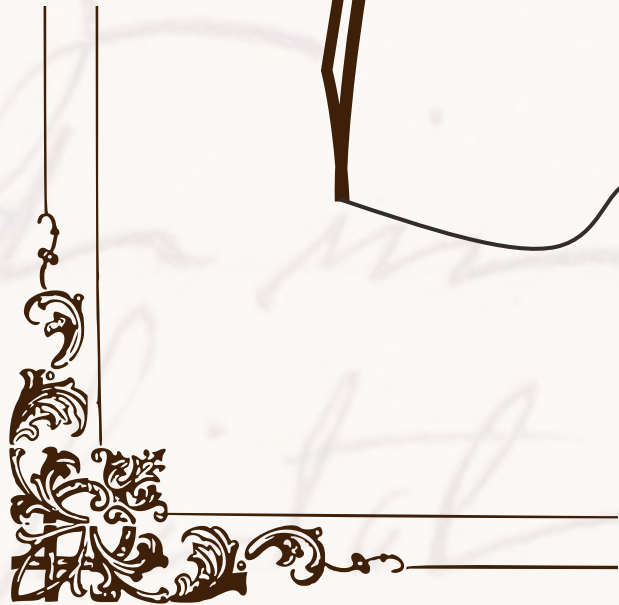




PROJECT PRESENTATION

AGENDA

01 Problem Statement

02 Importance of the Problem

03 Dataset Description

04 Solution Architecture

05 Exploratory Data Analysis

06 Inferential Stats

07 RFM Analysis

08 Suggested Solution

09 Algorithm and Methodology

10 Key Results and Insights

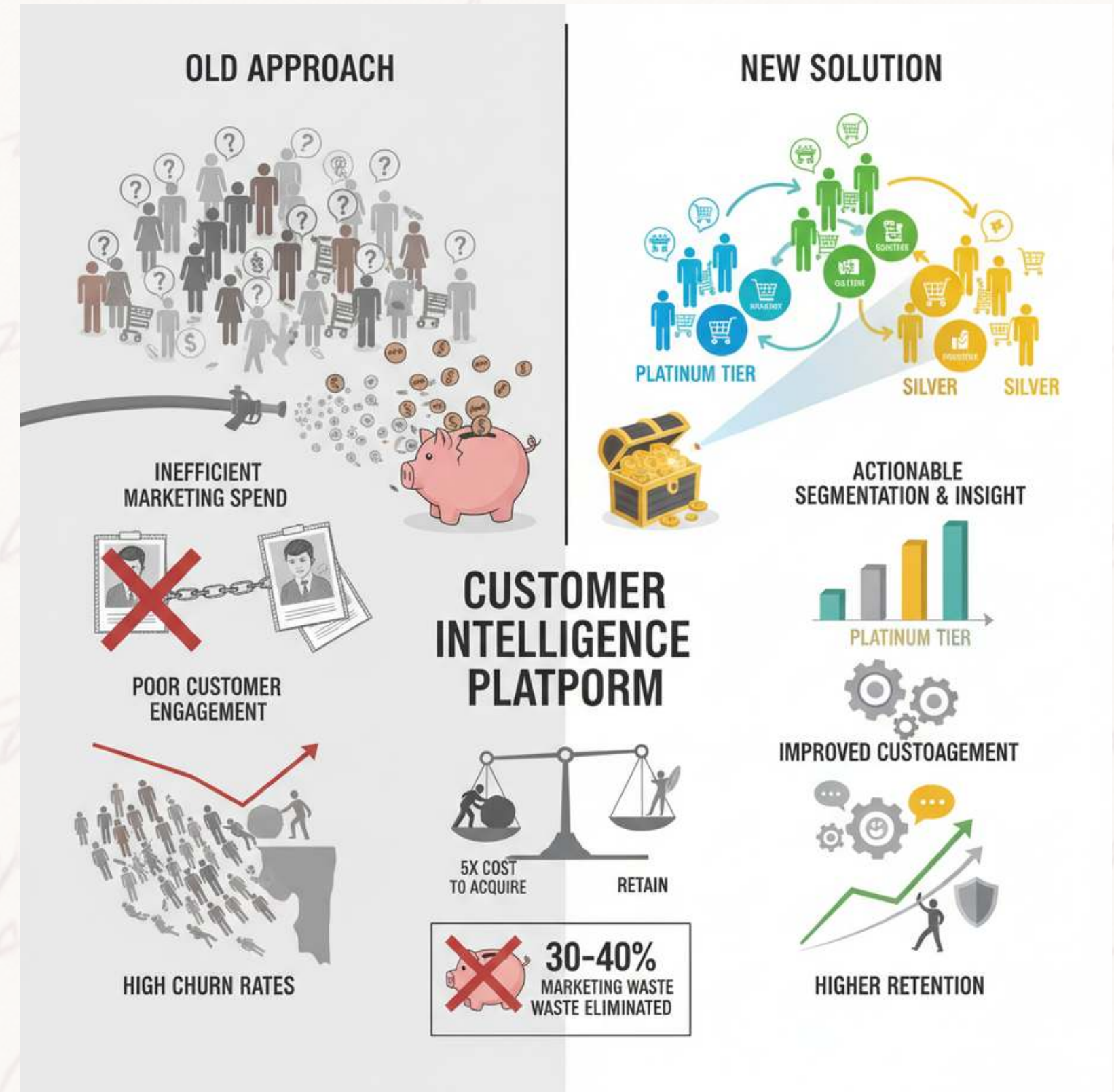


PROBLEM STATEMENT

Businesses struggle to identify and retain valuable customers due to lack of actionable segmentation and insights into purchasing behaviour leading to inefficient marketing spend and higher customer churn

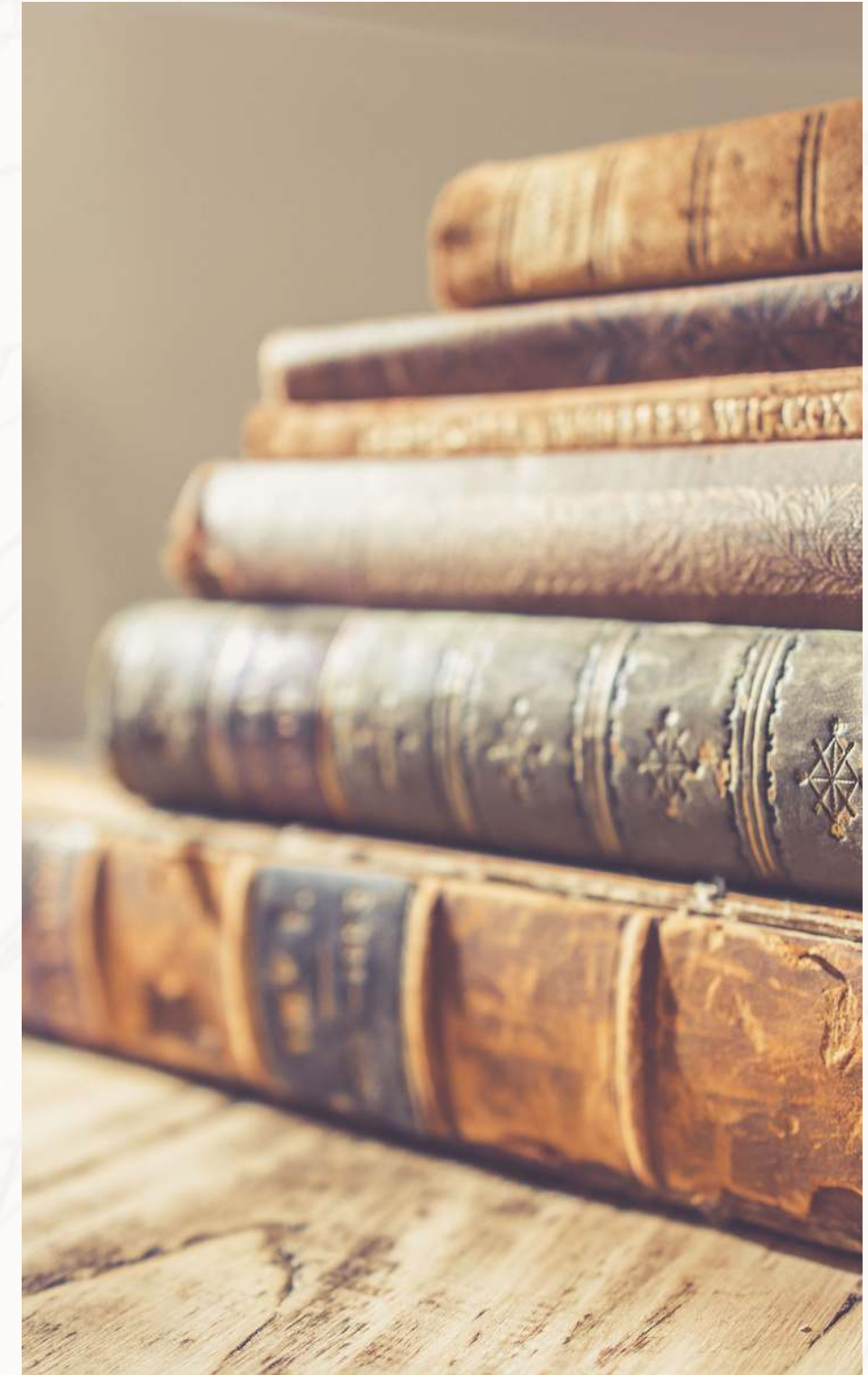
IMPORTANCE OF THE PROBLEM

- Customer retention directly drives long-term profitability.
- Loyal customers:
 - Contribute to steady revenue flow.
 - Are more likely to recommend the brand.
- Data-driven segmentation enables:
 - Targeted marketing
 - Better personalization
 - Enhanced lifetime value (CLV)



DATASET DESCRIPTION

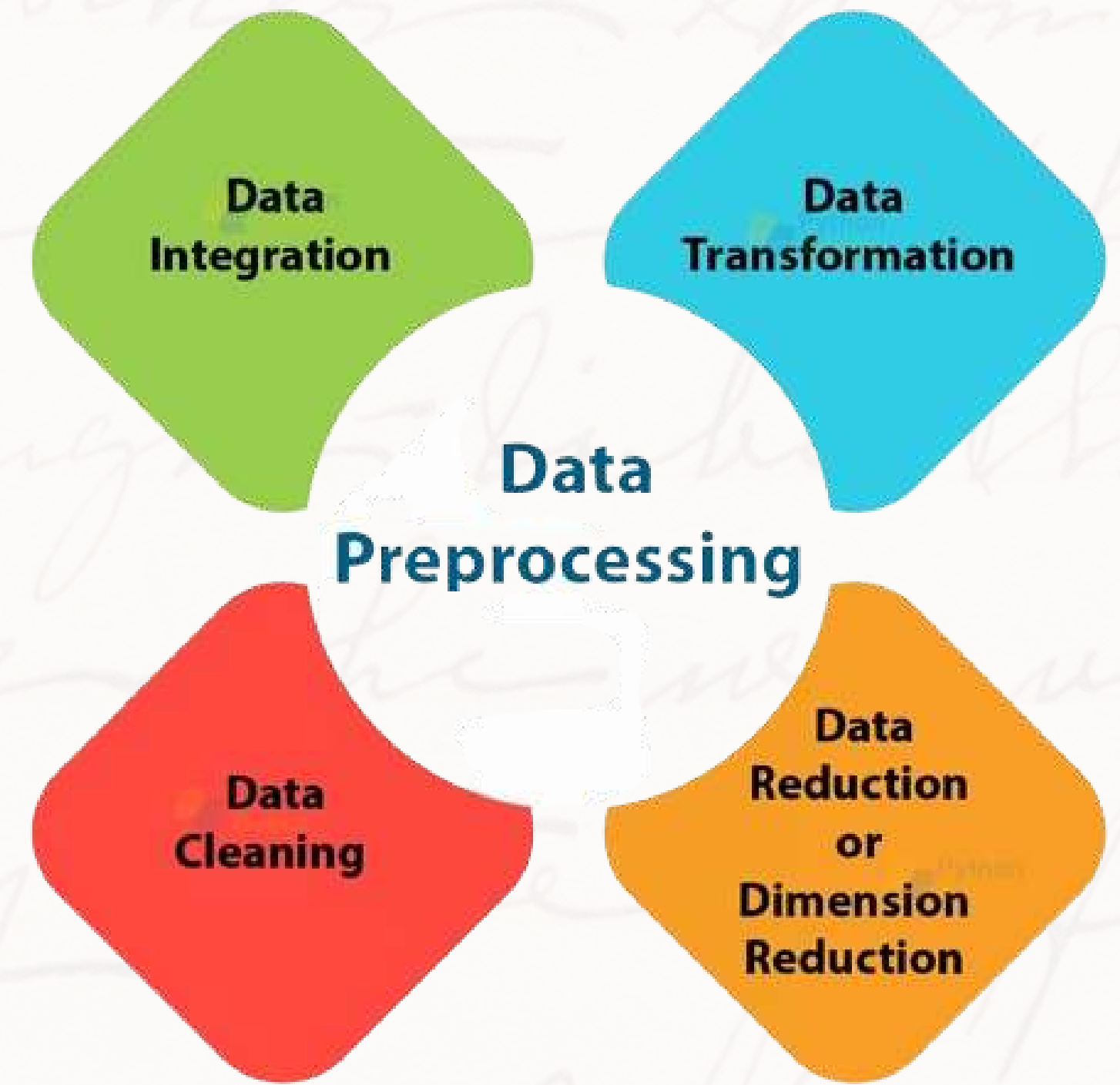
- **Total No. of Rows : ~300000**
- **Total No.of Columns: 30**
- **Numerical columns: 10**
- **Categorical columns: 20**
- **Dataset source : Kaggle**
- **Dataset Provider : Bhavikjikadara**
- **Dataset Name : Retail-transactional-dataset**
- **Target: Customer loyalty segmentation (via clustering)**



SOLUTION ARCHITECTURE

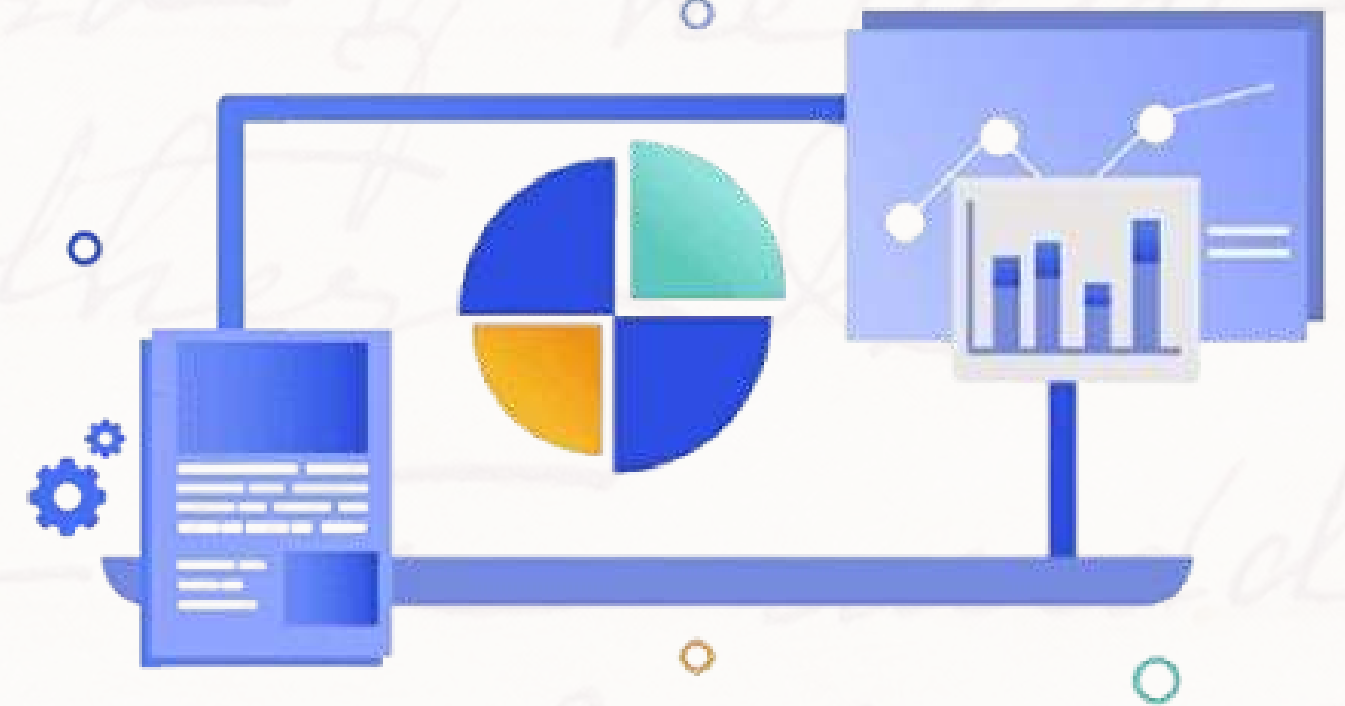
FUNCTIONAL FLOW

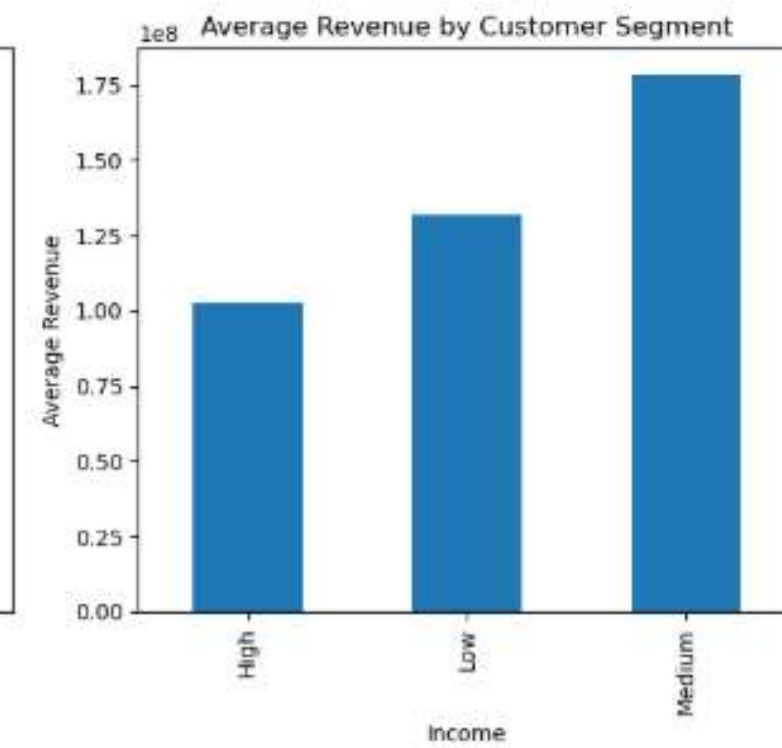
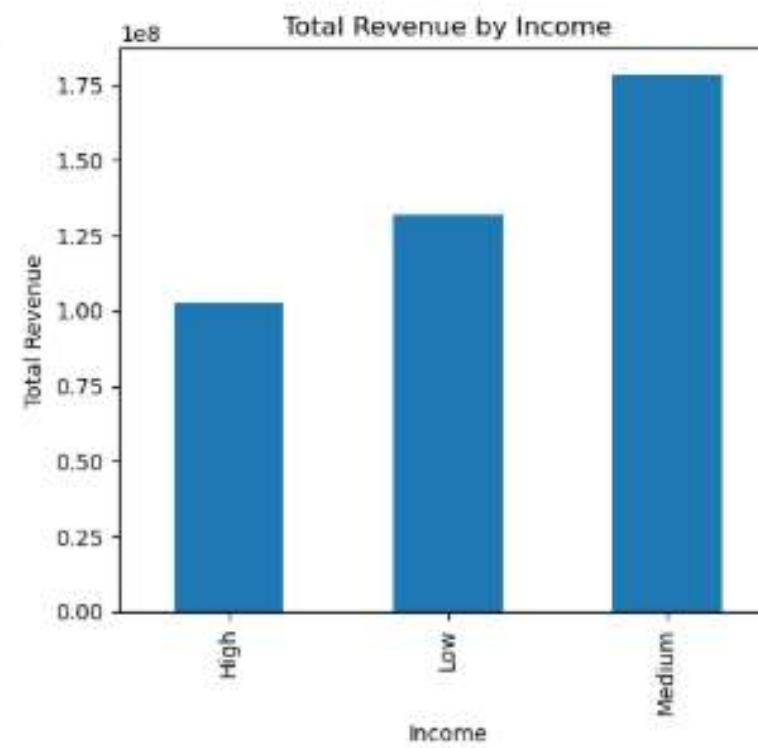
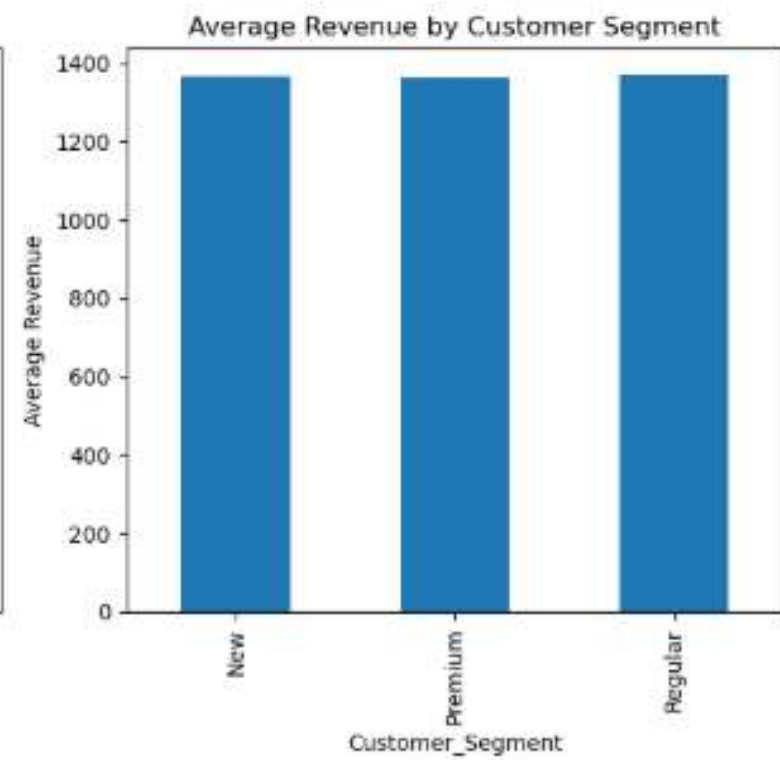
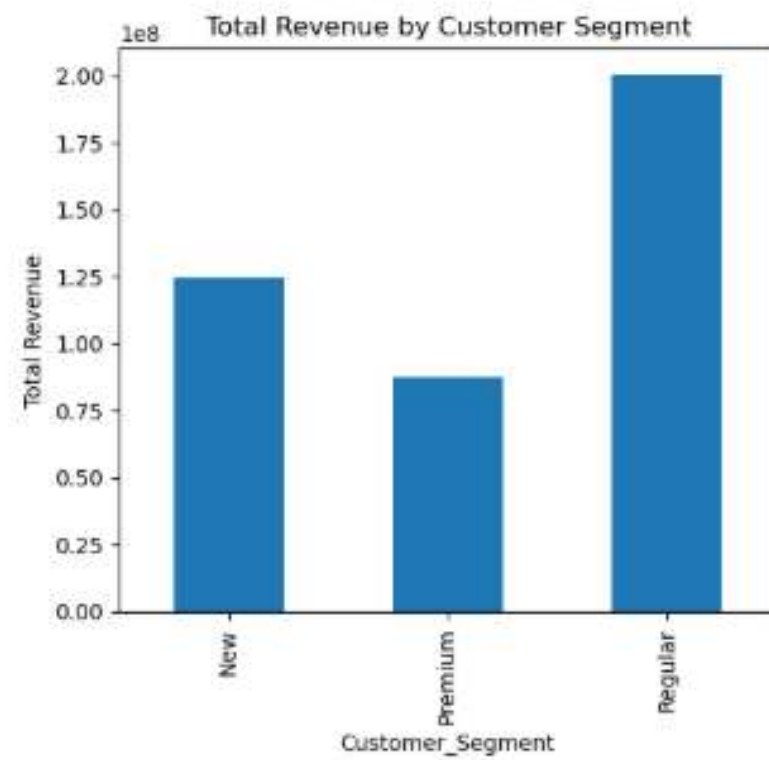
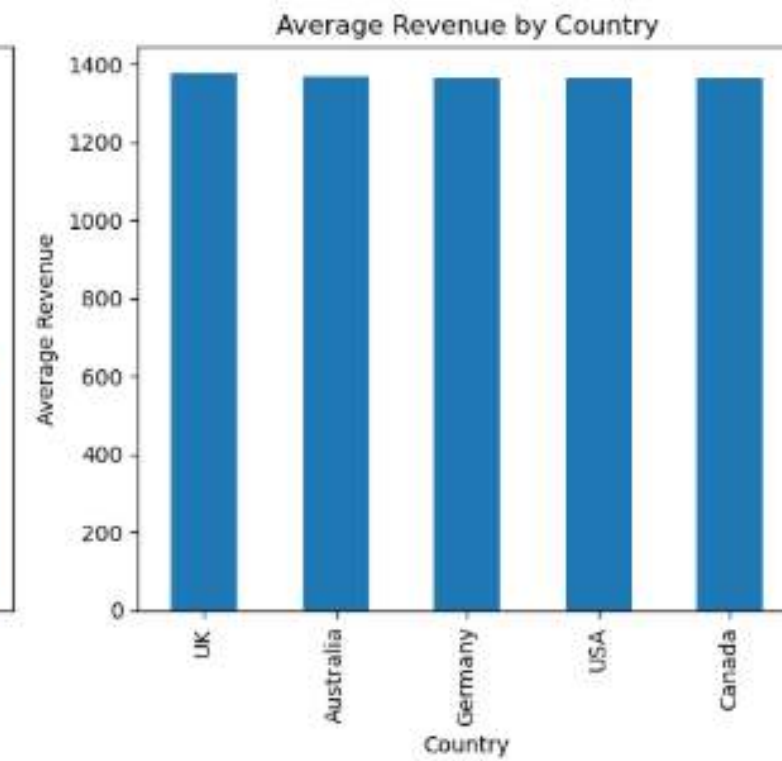
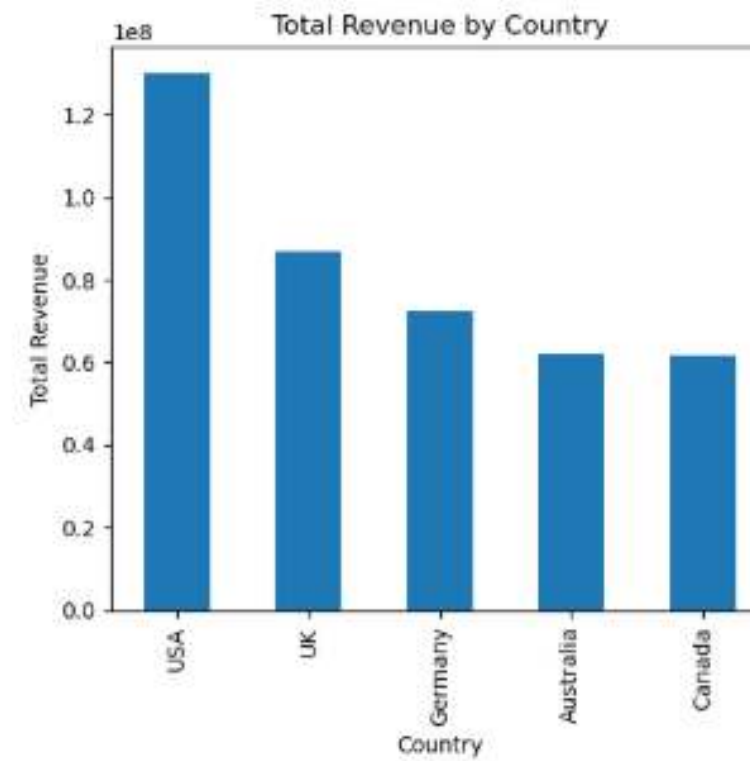
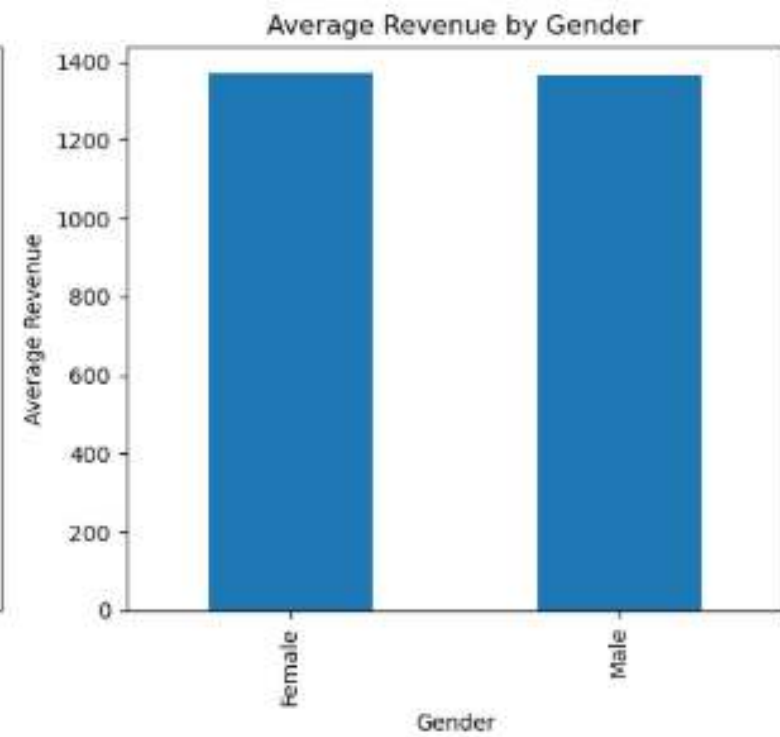
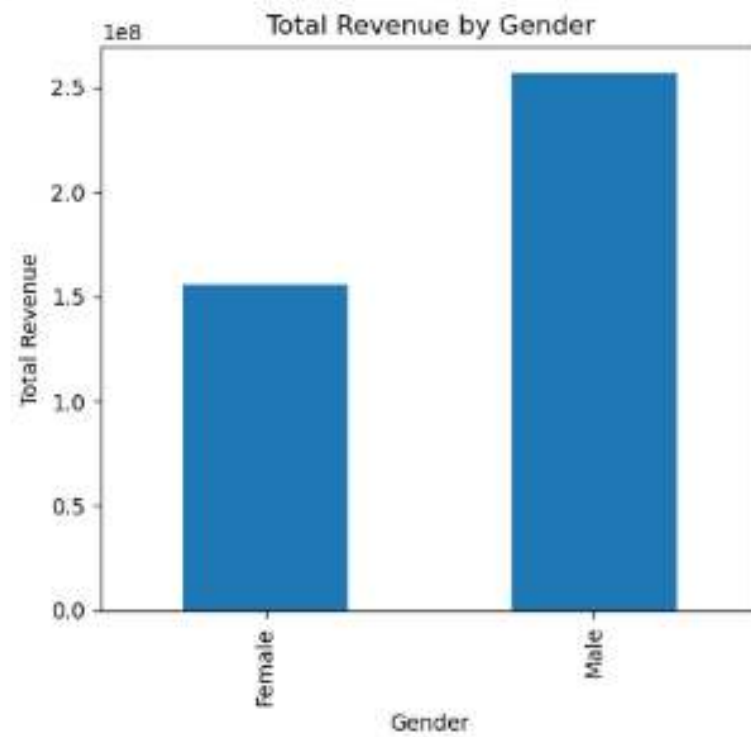
- 1. Data Input → 2. Preprocessing → 3. Clustering Model → 4. Insights
- Technical Stack:
 - Python (pandas, numpy, scikit-learn, matplotlib, seaborn)
 - Jupyter Notebook



EXPLORATORY DATA ANALYSIS (EDA)

- **Performed Tasks:**
 - **Missing value handling and Treating Duplicates**
 - **Visualizing Skewness**
 - **Correcting the datatype of Features**
 - **Data normalization for model compatibility**
 - **Visualization of sales trends & visit frequency**
- **Insights:**
 - **Removed 4 duplicate rows**
 - **Some customers show irregular purchase behavior → potential churners.**





INFERENCEAL STATS

Hypothesis 1: Is there a significant difference in spending between genders?

Hypothesis 2: Is there a relationship between Customer Segment and Feedback?

Hypothesis 3: Is there a difference in Ratings across Product Categories?

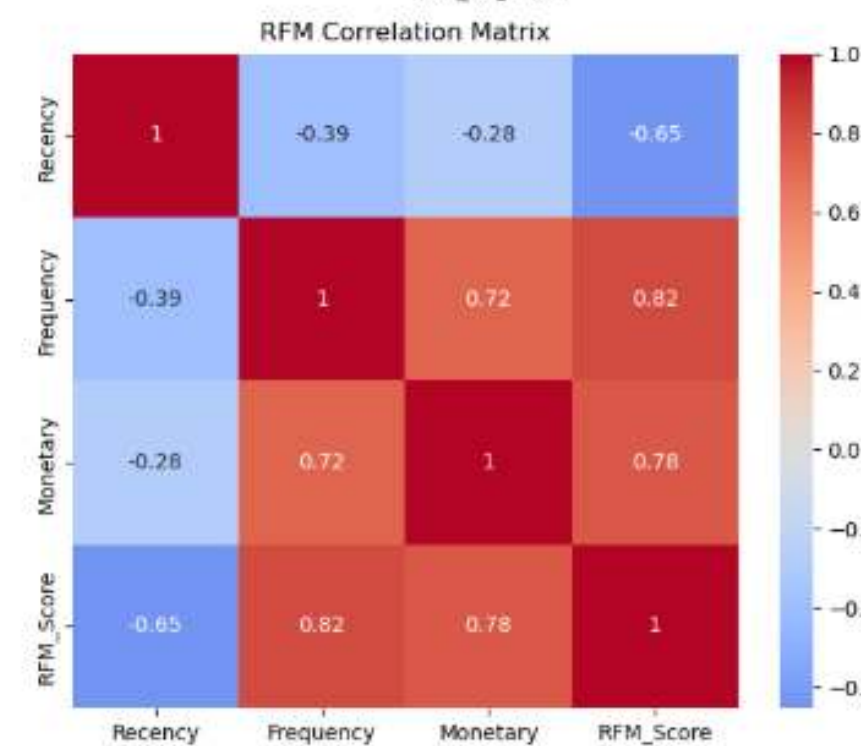
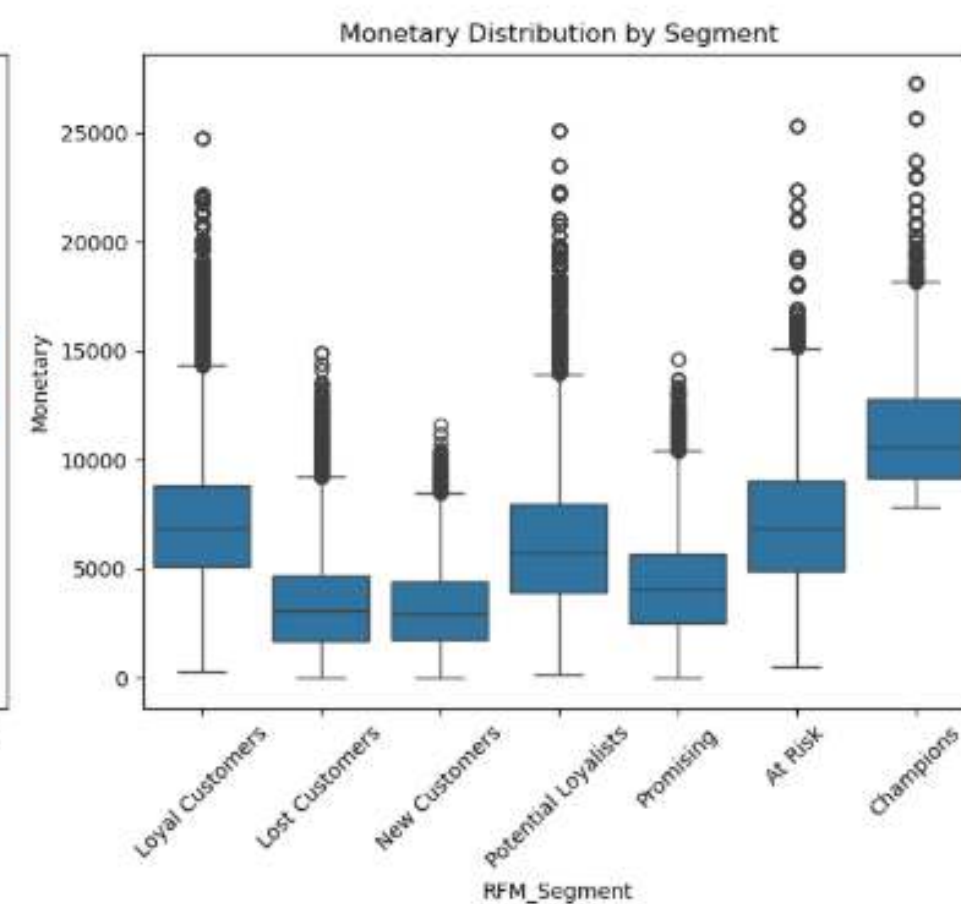
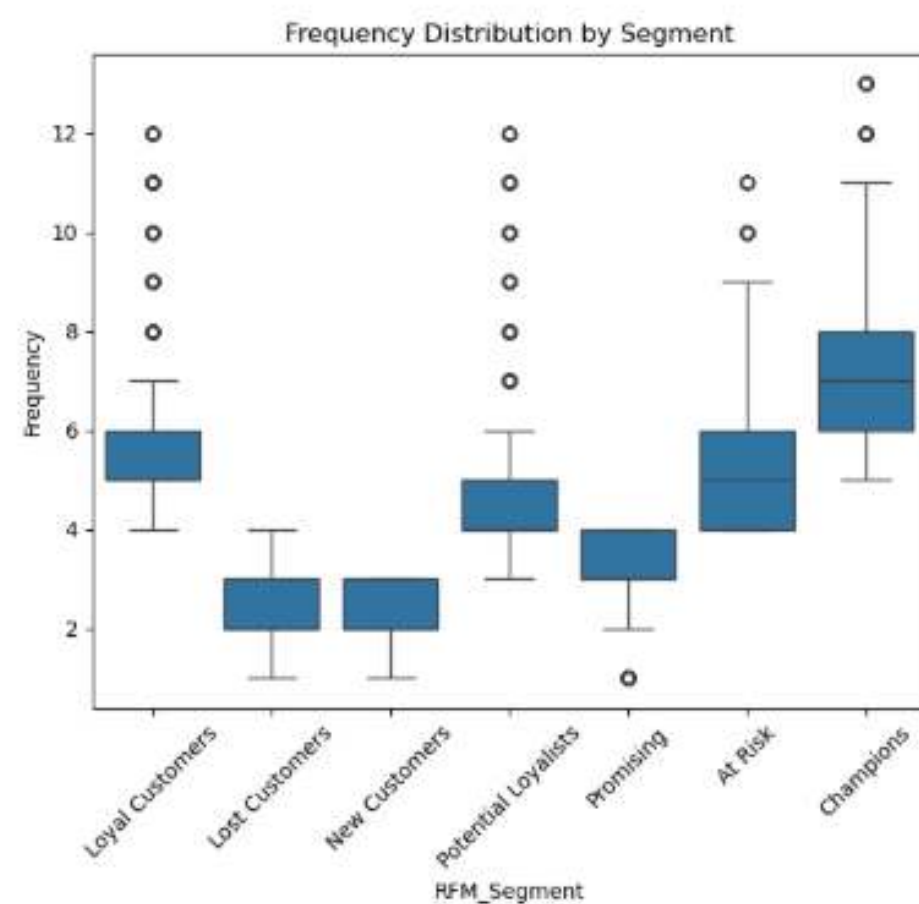
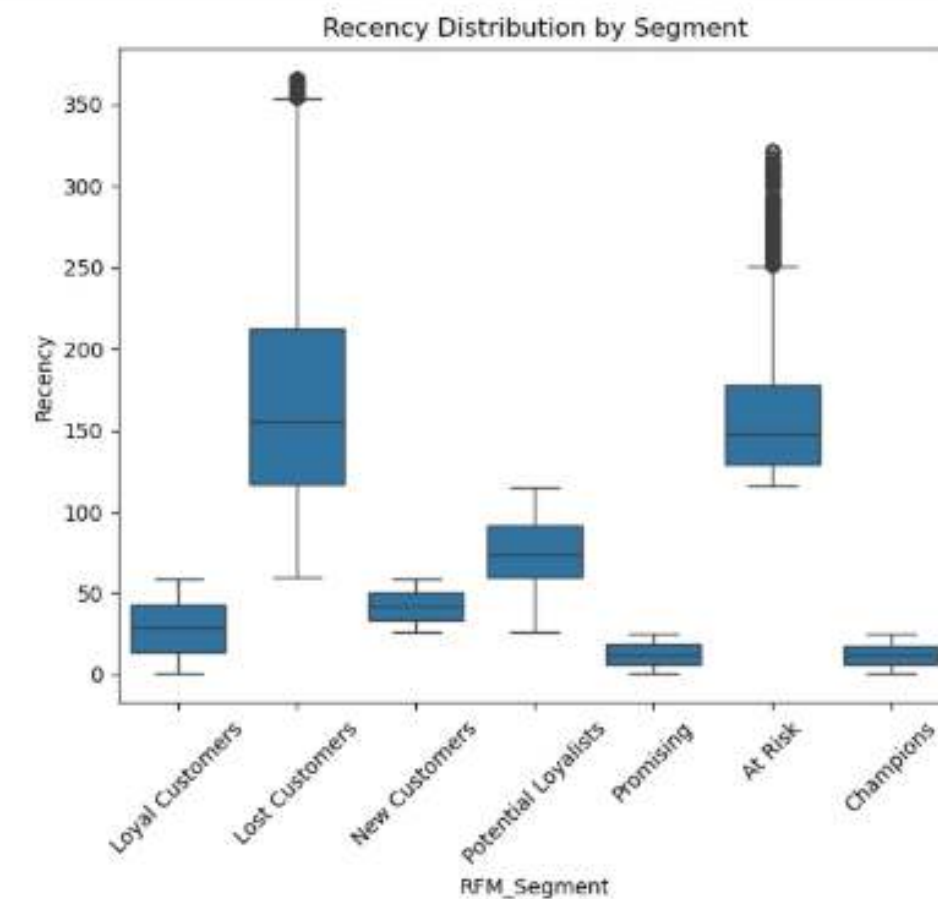
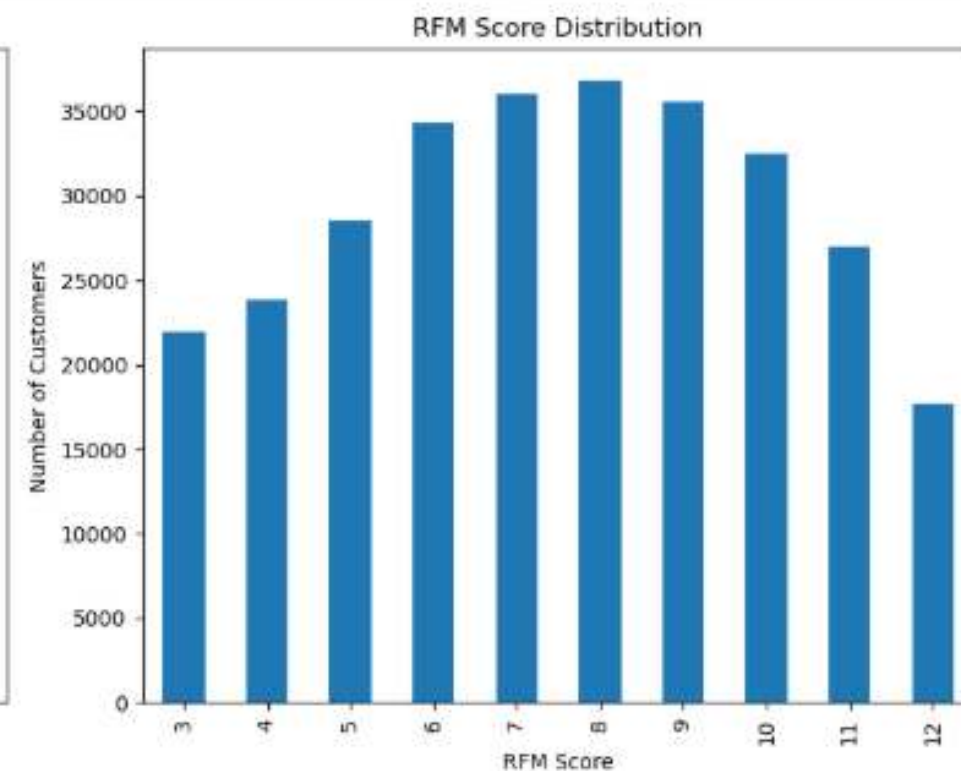
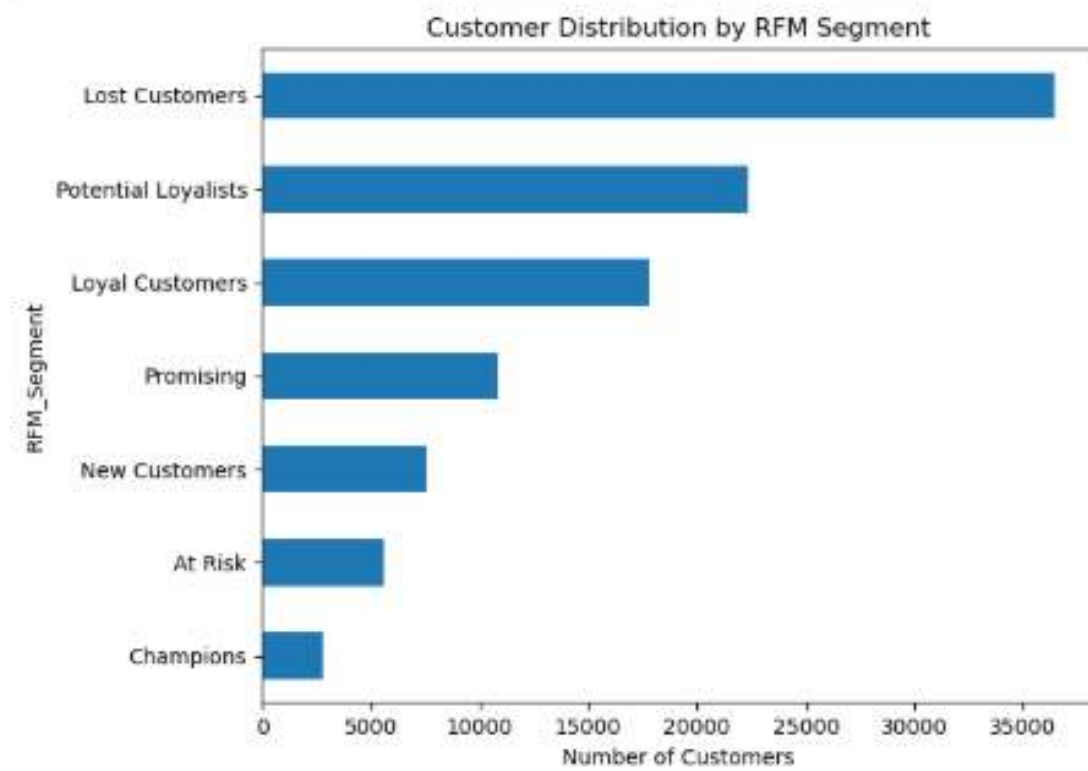
Hypothesis 4: Correlation between Age and Total Amount?

RFM ANALYSIS

- **Recency (R):** How recently a customer made a purchase.
 - More recent = more engaged
- **Frequency (F):** How often they purchase.
 - Frequent buyers show loyalty
- **Monetary (M):** How much they spend.
 - Higher spenders are more valuable

Purpose

- Identify loyal, potential, and at-risk customers.
- Support targeted marketing and personalized offers.
- Improve customer retention and lifetime value.



SUGGESTED SOLUTION

- **Use unsupervised machine learning for customer segmentation.**
- **Apply K-Means Clustering to group customers into loyalty tiers:**
 - **High-Value Loyal Customers**
 - **Potential Loyal Customers**
 - **Low-Engagement Customers**
- **Why K-Means?**
 - **Efficient for large numerical datasets.**
 - **Helps identify natural clusters in purchase patterns.**

ALGORITHM AND METHODOLOGY

- **Steps Implemented:**
 - **Data preprocessing (scaling, cleaning)**
 - **PCA for Dimension Reduction**
 - **Optimal K determination using Elbow Method**
 - **Apply K-Means Clustering**
 - **Label clusters and interpret profiles**
 - **Visualize clusters using PCA**

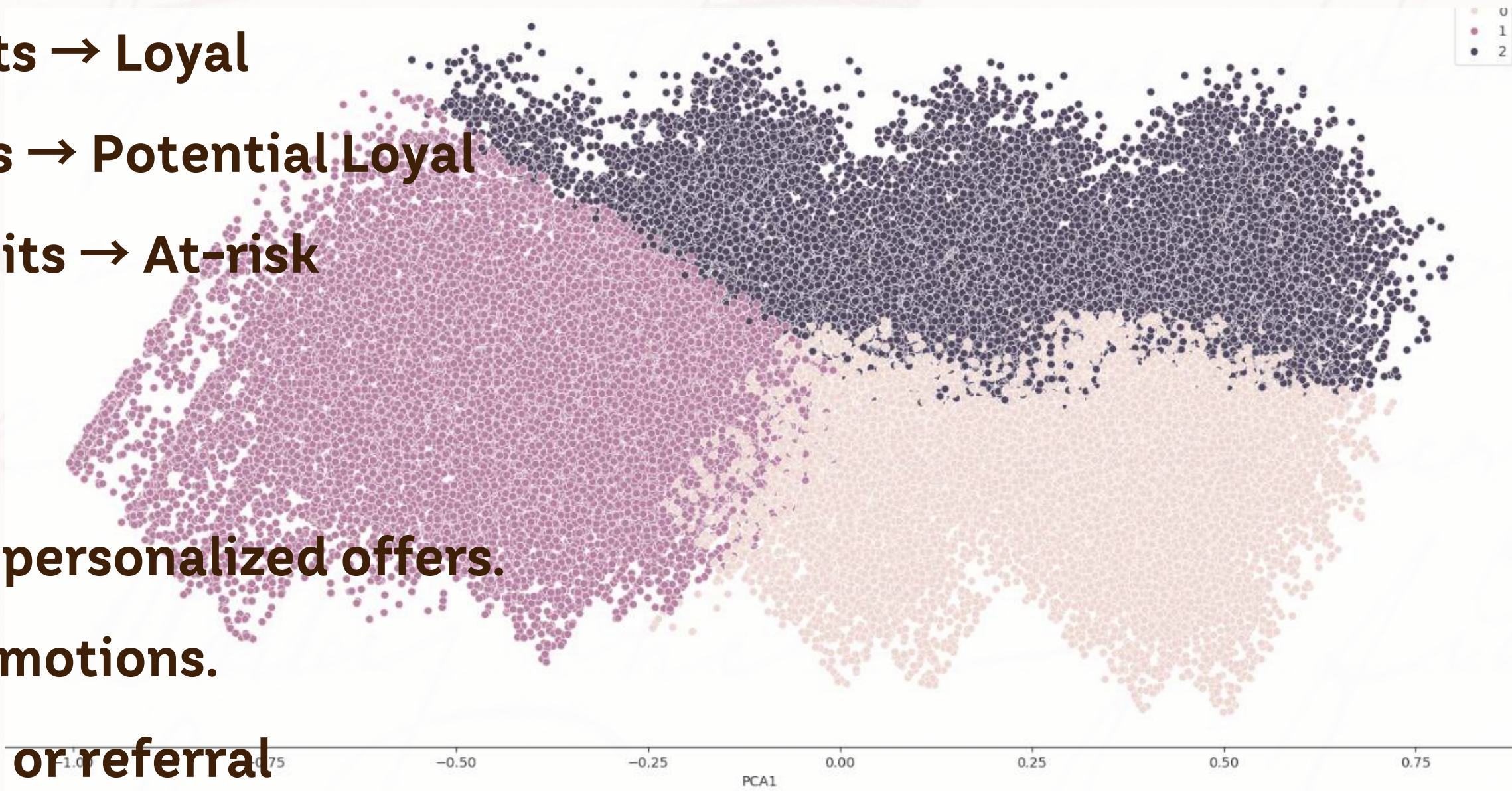
KEY RESULTS AND INSIGHTS

Cluster Profiles:

- **Cluster 0: High sales, frequent visits → Loyal**
- **Cluster 1: Moderate sales and visits → Potential Loyal**
- **Cluster 2: Low sales, infrequent visits → At-risk customers**

Business Insight:

- **Retain Cluster 0 with rewards and personalized offers.**
- **Engage Cluster 1 with targeted promotions.**
- **Re-engage Cluster 2 with discount or referral campaigns.**



A detailed illustration of a hand holding a quill pen. The hand is rendered with fine lines and shading, showing the fingers gripping the pen. The quill is long and slender, with a feathered end. The background features faint, cursive text, including the words "have to post" and "But", which are part of the larger text from the adjacent block.